A dense-case theorem for Seymour's second neighborhood conjecture

Jake Brukhman

CoinFund

jake@coinfund.io

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Abstract

Seymour's second neighborhood conjecture asserts that every finite oriented graph has a vertex with at least as many exact second outneighbors as outneighbors. Established cases include tournaments, proved by Fisher (1996), and oriented graphs of minimum outdegree at most six, proved by Kaneko and Locke (2001); a recent preprint of Sadhukhan, Sandeep, and Sen (2026) treats minimum outdegree seven. For dense incomplete graphs, Fidler and Yuster (2007) proved the conjecture when the missing edges form a matching, a star, or a clique, Ghazal (2012) extended this direction to generalized stars, and Dara, Francis, Jacob, and Narayanan (2022) proved it when the missing edges can be partitioned into a matching and a star. We give a short counting proof of the conjecture for every oriented graph of order $n = 2\delta + 2$, where δ is the minimum outdegree, with no prescribed structure on the missing edges. Together with Fisher's tournament theorem, this implies the conjecture for every oriented graph satisfying $n \leq 2\delta + 2$. Combined with the known minimum-outdegree results, this raises the best lower bound known to us on the order of a counterexample from 16 to 17 and, conditional on the preprint of Sadhukhan, Sandeep, and Sen (2026), from 18 to 19.

1 Introduction

A finite *directed graph* is a pair $G = (V(G), A(G))$, where $V(G)$ is a finite set of vertices and $A(G)$ is a set of ordered pairs of distinct vertices, called *arcs*. We write

$$u \to v$$

when $(u, v) \in A(G)$. Thus the arrow points from the source of the arc to its target. The directed graph is *oriented* if, for every two distinct vertices u, v, at most one of $u \to v$ and $v \to u$ is present. For a vertex $v \in V(G)$, define its outdegree and indegree by

$$d^+(v) = |\{x \in V(G) : v \to x\}|, \qquad d^-(v) = |\{x \in V(G) : x \to v\}|.$$

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The minimum outdegree of G is $\delta = \min_{v \in V(G)} d^+(v)$. For a vertex $v \in V(G)$, define the *outneighborhood* by

$$N_1(v) = \{x \in V(G) : v \to x\},$$

and the *exact second outneighborhood* by

$$N_2(v) = \{x \in V(G) \setminus N_1(v) : y \to x \text{ for some } y \in N_1(v)\}.$$

By definition $N_2(v)$ is disjoint from $N_1(v)$; since G is oriented, it also excludes v. Thus $N_2(v)$ consists exactly of the vertices reached from v in two steps but not in one step. When discussing reachability from v to x, we call v the *source vertex* and x the *target vertex*. A vertex v is a *Seymour vertex* if $|N_2(v)| \geq |N_1(v)|$.

Conjecture (Seymour)**.** *Every finite oriented graph has a Seymour vertex.*

Several results bear directly on dense or extremal cases. Fidler and Yuster [5] proved the conjecture for several prescribed missing-edge structures, and Ghazal [7] generalized that line of work. Dara, Francis, Jacob, and Narayanan [3] proved the conjecture when the missing edges can be partitioned into a matching and a star. Further prescribed missing structures, including two stars and disjoint paths, were treated by Daamouch, Ghazal, and Al-Mniny [2], and Daamouch [1] proved the conjecture for related digraph classes. Espuny Díaz, Girão, Granet, and Kronenberg [4] obtained reductions involving minimum outdegree, and Guo, Kang, and Zwaneveld [8] studied Seymour-tight orientations and counterexamples close to regular tournaments.

Let G be an oriented graph on n vertices with minimum outdegree δ. Counting arcs by their sources gives

$$n\delta \leq |A(G)| \leq \binom{n}{2},$$

so $n \geq 2\delta + 1$. Equality forces every pair of vertices to be adjacent and every outdegree to equal δ, so the graph is a regular tournament, which satisfies the conjecture by Fisher's theorem [6].

Theorem 1.1 (Dense-case theorem)**.** *Let G be an oriented graph on n vertices with minimum outdegree δ. If*

$$n = 2\delta + 2,$$

then G has a Seymour vertex.

At $n = 2\delta + 2$, up to $n/2$ edges may be missing and no structure is imposed on them, in contrast to [1, 2, 3, 5, 7]; conversely, those results impose no lower bound on the minimum outdegree, so neither result contains the other.

Corollary 1.2 shows that any counterexample to the conjecture must satisfy $n \geq 2\delta+3$: no counterexample can occur at the two densest orders an oriented graph of minimum outdegree δ can have. This complements the study of counterexamples close to regular tournaments in [8].

Corollary 1.2. *Let G be an oriented graph on n vertices with minimum outdegree δ. If $n \leq 2\delta + 2$, then G has a Seymour vertex.*

Proof. Every oriented graph satisfies $n \geq 2\delta+1$. If $n = 2\delta+1$, then G is a regular tournament as observed above, and Fisher's theorem [6] applies. If $n = 2\delta + 2$, apply Theorem 1.1. $\square$

Corollary 1.3 (Minimum counterexample order)**.** *Every counterexample to Seymour's conjecture has at least* 17 *vertices. Assuming the result of Sadhukhan et al. [10], every counterexample has at least* 19 *vertices.*

Proof. By Kaneko and Locke's theorem [9], a counterexample has minimum outdegree $\delta \geq 7$. Corollary 1.2 then gives $n \geq 2\delta + 3 \geq 17$. Under the result of Sadhukhan, Sandeep, and Sen, the case $\delta = 7$ is also excluded, so $\delta \geq 8$ and hence $n \geq 19$. $\square$

Without Theorem 1.1, the same reasoning gives only $n \geq 2\delta + 2$, since Fisher's theorem excludes just the order $n = 2\delta + 1$; that is, $n \geq 16$ unconditionally and $n \geq 18$ conditionally. To our knowledge, no stronger order bounds were previously known.

2 The fixed-target argument

For a source vertex v, define

$$U(v) = V(G) \setminus \Big(\{v\} \cup N_1(v) \cup N_2(v)\Big).$$

Thus $U(v)$ is the set of target vertices that the source v cannot reach in at most two steps. Now fix a target vertex x. Define the *trap* of x by

$$T(x) = \{y \in V(G) \setminus \{x\} : y \nrightarrow x\}.$$

These are exactly the vertices that do not send an arc to x. Because G is oriented, every outneighbor of x belongs to $T(x)$. Hence

$$|T(x)| \geq d^+(x) \geq \delta.$$

Define the *target slack*

$$q(x) = |T(x)| - \delta \geq 0.$$

Now define

$$P(x) = \{v \in V(G) : x \in U(v)\}.$$

Thus $P(x)$ is the set of source vertices from which the fixed target x cannot be reached in at most two steps. If $v \in P(x)$, then $v \nrightarrow x$. Moreover, no vertex of $N_1(v)$ points to x, since otherwise there would be a two-step path from v to x. Therefore

$$\{v\} \cup N_1(v) \subseteq T(x). \tag{1}$$

This is the fixed-target observation: every source v that cannot reach x in 2 steps, together with all of its outneighbors, is confined to the same trap $T(x)$.

Lemma 2.1 (Fixed-target capacity)**.** *Let x be a target vertex. If $P(x) \neq \varnothing$, then*

$$q(x) \geq 1 \qquad \text{and} \qquad |P(x)| \leq 2q(x) - 1.$$

In particular, if $q(x) = 0$, then $P(x) = \varnothing$.

Proof. Suppose $P(x) \neq \varnothing$, and set $p := |P(x)| \geq 1$. By (1), $P(x) \subseteq T(x)$, and every arc whose source lies in $P(x)$ has its target in $T(x)$. There are at least δp such arcs. At most $\binom{p}{2}$ have both endpoints in $P(x)$, because G is oriented, and at most $p\big(|T(x)| - p\big)$ go from $P(x)$ to the rest of the trap. Consequently,

$$\delta p \leq \binom{p}{2} + p\big(\delta + q(x) - p\big).$$

Dividing by p and rearranging gives $p \leq 2q(x) - 1$. Since $p \geq 1$, this forces $q(x) \geq 1$. The final assertion is the contrapositive. $\square$

Proof of Theorem 1.1. Suppose for contradiction that G has no Seymour vertex. Then $\delta > 0$, since a vertex of outdegree zero is automatically a Seymour vertex. Let

$$L = \{v \in V(G) : d^+(v) = \delta\}, \qquad \ell = |L| \geq 1.$$

Every $v \in L$ is not a Seymour vertex, so $|N_2(v)| \leq \delta - 1$. Since $n = 2\delta + 2$, it follows that

$$|U(v)| = n - 1 - \delta - |N_2(v)| \geq 2. \tag{2}$$

Thus each minimum-outdegree source contributes at least two unreachable targets. Let

$$I = \Big|\big\{(v, x) \in V(G)^2 : x \in U(v)\big\}\Big|.$$

Counting these ordered source–target pairs first by their source and then by their target gives

$$I = \sum_{v \in V(G)} |U(v)| = \sum_{x \in V(G)} |P(x)| \tag{3}$$

since $x \in U(v)$ and $v \in P(x)$ describe the same ordered pair (v, x). From (2),

$$I \geq 2\ell > 0. \tag{4}$$

Put

$$Q = \sum_{x \in V(G)} q(x).$$

Since $I > 0$, at least one set $P(x)$ is nonempty; Lemma 2.1 then implies that at least one $q(x)$ is positive. Applying the lemma in (3) gives

$$I \leq \sum_{q(x) > 0} \big(2q(x) - 1\big) < 2 \sum_{x} q(x) = 2Q. \tag{5}$$

It remains to compare Q with ℓ. Since $|T(x)| = n - 1 - d^-(x)$,

$$q(x) = n - 1 - \delta - d^-(x).$$

Total indegree equals total outdegree, and every vertex outside L has outdegree at least $\delta + 1$. Therefore

$$\begin{aligned} Q &= n(n - 1 - \delta) - \sum_{x} d^+(x) \\ &\leq n(n - 1 - \delta) - \big(\ell\delta + (n - \ell)(\delta + 1)\big) \\ &= \ell, \end{aligned} \tag{6}$$

where the last equality uses $n = 2\delta + 2$. Now (4)–(6) give

$$I < 2Q \leq 2\ell \leq I,$$

a contradiction. Hence G has a Seymour vertex. $\square$

Remark. The order $n = 2\delta+2$ also represents the limitation of this argument. For $n = 2\delta+3$, the same computation as in (6) gives only $Q \leq \ell + n$, while (2) gives $I \geq 3\ell$; since $\ell \leq n$, the comparison $I < 2Q \leq 2\ell + 2n$ no longer yields a contradiction. Extending the method past $n = 2\delta + 2$ therefore requires new ideas.

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